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Student Dropout Risk Prediction Using Machine Learning

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ABSTRACT

Student dropout is a critical issue affecting educational institutions worldwide, leading to loss of talent and reduced institutional efficiency. Existing systems often fail to provide early and accurate identification of at-risk students due to limited feature usage and lack of real-time prediction capabilities.

In this work, a machine learning-based **Dropout Risk Detection System** is proposed, which leverages academic, demographic, and financial features to classify students as *At-Risk* or *Not At-Risk*. Multiple models including Logistic Regression, Random Forest, and Gradient Boosting are trained and evaluated. The final system selects Gradient Boosting as the optimal model, achieving an accuracy of **88.70%** and F1-score of **0.8214**. The system also provides risk levels, reasons, and suggestions, making it interpretable and actionable. This work contributes a lightweight, practical, and student-centric predictive framework for early intervention.

Keywords: Machine Learning, Dropout Prediction, Gradient Boosting, SMOTE, Classification

I. INTRODUCTION

A. Background

Student dropout is a persistent challenge in higher education systems. Factors such as poor academic performance, financial instability, and lack of engagement significantly contribute to dropout rates. Traditional monitoring systems are reactive rather than proactive, identifying issues only after students have already disengaged.

With the rise of machine learning, predictive models can now analyze historical student data to identify patterns associated with dropout risk, enabling early intervention.

B. Importance and SDG Link

This work aligns with:

- **SDG 4: Quality Education** – Improving student retention
- **SDG 10: Reduced Inequalities** – Supporting vulnerable students
- **SDG 9: Innovation in Education** – Applying AI-driven solutions

By identifying at-risk students early, institutions can take corrective actions such as counseling, financial aid, or academic support.

C. Problem Statement and Objectives

Problem Statement:
Existing dropout prediction systems suffer from limited feature diversity, lack of real-time usability, and poor generalization.

Objectives:

- Build an accurate ML model to predict dropout risk
- Compare multiple algorithms and select the best model
- Identify key factors influencing dropout
- Provide early alerts with actionable insights

D. Novelty of Proposed Work

The proposed system introduces:

- Use of **multi-dimensional features** (academic + financial + demographic)
- **SMOTE-based imbalance handling** for better model performance
- **Risk-level classification (Low, Medium, High)** instead of binary output
- **Explainable predictions** with reasons and suggestions
- **User-input based prediction system** for real-world usability

II. LITERATURE REVIEW

Several studies have explored dropout prediction using machine learning:

- Villar & Andrade (2024) used multiple ML models but lacked real-time deployment
- Marcolino et al. (2025) focused on LMS logs but ignored socio-economic factors
- Cho et al. (2023) relied only on academic performance
- Kim et al. (2023) proposed a complex hybrid model with limited generalization

TABLE I — Comparison of Related Work

Ref	Author	Problem	Dataset	Method	Features
[1]	Villar & Andrade, 2024	Limited to single dataset, lacks real-time deployment	Real-world university dataset	Multiple ML models	Academic + student records
[2]	Marcolino et al., 2025	Depends only on LMS logs, ignores socio-economic factors	LMS / Moodle logs	CatBoost + NSGA-II	Clickstream, engagement
[3]	Cho et al., 2023	Focuses only on academic performance, missing emotional/behavioral factors	Large academic dataset	Multi-model + SMOTE	Grades, attendance
[4]	Kim et al., 2023	Complex hybrid model, difficult to generalize across universities	Single university dataset	Hybrid ML + SHAP	Academic + clustering features

Research Gap

Existing systems:

- Use limited datasets
- Ignore multi-factor influence
- Lack real-time usability
- Are complex and not interpretable

Proposed Contribution

This work addresses these gaps by:

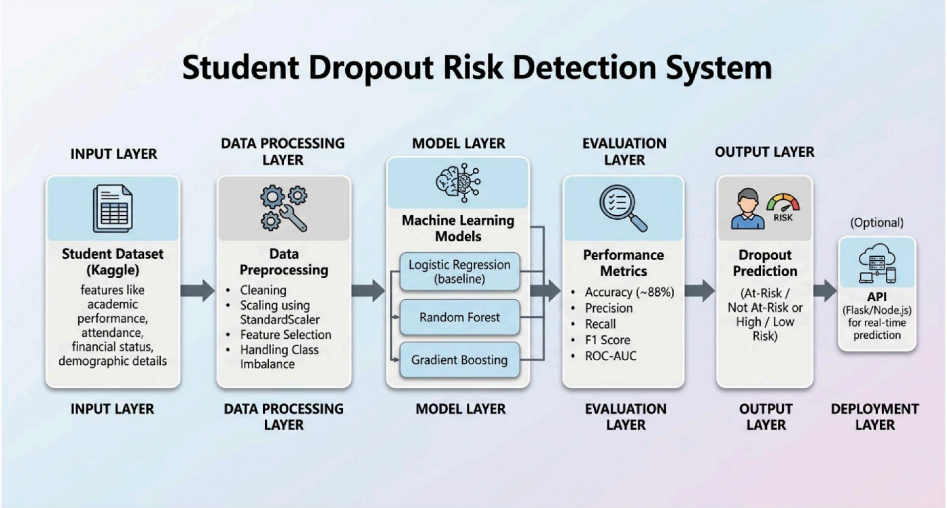
- Combining **academic + financial + engagement features**
- Using **simple yet effective ML models**
- Providing **interpretable and user-friendly outputs**

III. METHODOLOGY

A. System Architecture

The proposed **Student Dropout Risk Detection System** follows a structured multi-layer pipeline as illustrated in the methodology diagram. The system is divided into five major layers: Input Layer, Data Processing Layer, Model Layer, Evaluation Layer, and Output Layer. An optional deployment layer is also included for real-time usage.

The workflow ensures a systematic transformation of raw student data into actionable predictions.



B. Dataset Description

The dataset contains student-related information including:

1. Demographic Features

- Age at enrollment
- Gender
- Nationality

2. Socio-Economic Features

- Scholarship status
- Debtor status
- Tuition fee payment

3. Academic Features

- 1st semester performance
- 2nd semester performance

4. Macro-Economic Indicators

- Inflation rate
- GDP
- Unemployment rate

C. Target Variable

Original classes:

- Dropout
- Graduate
- Enrolled

Converted into:

- **1 → Dropout**
- **0 → Not Dropout**

D. Data Preprocessing

The preprocessing stage plays a critical role in transforming raw student data into a structured format suitable for machine learning. The proposed system follows a **five-step preprocessing pipeline.**

Step 1: Dataset Mean Initialization

The system begins by loading historical student data and computing statistical measures such as mean values for numerical features.

- Missing values are handled using mean imputation
- Ensures dataset consistency and completeness
- Provides a stable baseline for real-time input handling

This step ensures that incomplete records do not negatively affect model performance

Step 2: Feature Updating

The system updates key student attributes dynamically:

- Age at enrollment
- Scholarship status
- Debtor status
- Tuition fee payment
- Semester performance

This step ensures that the model uses latest and most relevant data, enabling real-time prediction.

Step 3: Feature Alignment and Structuring

All processed features are combined into a standardized dataset format:

- Feature ordering is maintained
- Data types are aligned
- Missing values are filled
- Structured into a model-ready format

This ensures consistency between training data and real-time input data.

Step 4: Model Prediction Preparation

The structured dataset is then passed to the trained machine learning model.

- Data is scaled (StandardScaler)
- Ensures compatibility with trained model distribution

- Prevents bias due to feature magnitude differences

Step 5: Risk Classification

The final output is categorized into risk levels:

- Low Risk
- Medium Risk
- High Risk

This classification improves interpretability and helps institutions take targeted actions.

E. Feature Selection

Selected features:

- Age at enrollment
- Scholarship holder
- Debtor
- Tuition fees status
- 1st semester approved units
- 2nd semester approved units

These features were chosen based on importance and practical usability.

F. Model Training

Three machine learning models were implemented:

1. Logistic Regression

- Simple and interpretable
- Good baseline model

2. Random Forest

- Ensemble model
- Handles non-linearity

3. Gradient Boosting

- Sequential learning
- High accuracy and performance

G. Model Pipeline

Steps:

1. Train-test split
2. Apply SMOTE
3. Train models
4. Evaluate performance
5. Select best model

H. Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score

IV. RESULTS AND DISCUSSION

A. Model Performance

TABLE II — Comparison Of Models

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.8746	0.7836	0.8415	0.8115
Random Forest	0.8825	0.8409	0.7817	0.8102
Gradient Boosting	0.8870	0.8333	0.8099	0.8214

B. Best Model Selection

- Selected Model: Gradient Boosting
- Accuracy: 0.8870
- Precision: 0.8333
- Recall: 0.8099
- F1 Score: 0.8214

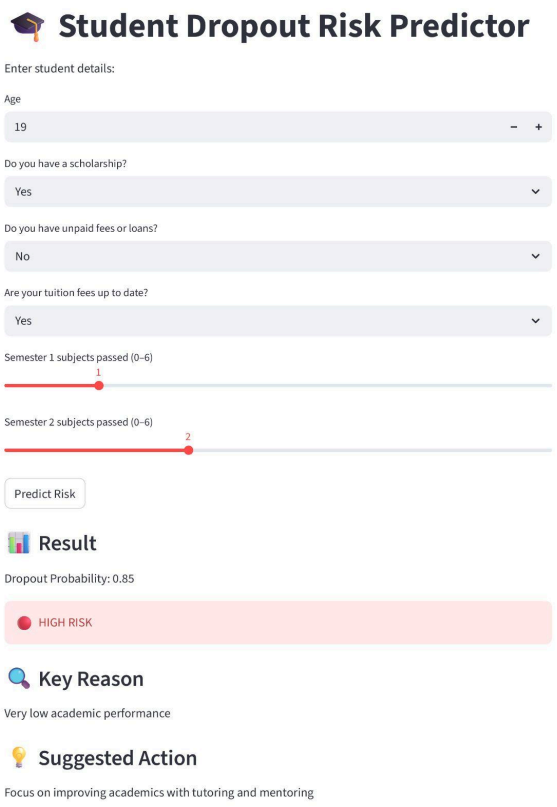
Reason for selection

- Achieved the highest F1-score, giving the best balance between precision and recall
- Provided overall best performance among all models
- More reliable for detecting dropout cases without too many false predictions

C. System Output

The system provides:

- Prediction (Dropout / Not Dropout)
- Risk Level (Low / Medium / High)
- Reason for risk
- Suggested actions



D. Key Observations

- Data preprocessing significantly improves performance by cleaning the dataset, standardizing feature values, and ensuring that the machine learning models receive consistent and reliable input. This reduces noise in the data and helps

the models learn meaningful patterns more effectively.

- SMOTE helps in handling class imbalance by generating synthetic samples for the minority class, which in this case represents dropout cases. Without this step, the model could become biased toward the majority class and fail to detect at-risk students accurately.
- Feature selection reduces complexity by removing irrelevant or less important attributes from the dataset. This not only improves computational efficiency but also helps the model focus on the most influential factors affecting dropout risk, leading to better generalization.
- Gradient Boosting performs best overall because it combines the predictions of multiple weak learners in a sequential manner, gradually correcting previous errors. This allows it to capture complex relationships in the data and achieve the best balance between precision, recall, and F1-score among the tested models.

V. CONCLUSION

This research presents a machine learning-based system for predicting student dropout risk by analyzing academic, demographic, and socio-economic factors. The proposed approach demonstrates that dropout prediction can be effectively treated as a classification problem, where historical student data is used to identify patterns associated with academic failure or disengagement. By combining preprocessing techniques, feature selection, imbalance handling, and model comparison, the system is able to generate reliable predictions that can support early intervention.

Among the evaluated models, Gradient Boosting emerged as the most effective due to its strong balance between precision, recall, and F1-score. This indicates that the model is not only capable of identifying students who are truly at risk, but also does so with fewer incorrect predictions. The use of SMOTE further improved the model's ability to learn from the minority dropout class, which is essential in educational datasets where dropout cases are often less frequent than non-dropout cases. In addition, the inclusion of interpretable outputs such as risk level, reason, and suggestions makes the system more practical for institutional use.

A major strength of the proposed system is that it does not simply provide a binary prediction. Instead, it translates model output into actionable information that can help educators, counselors, and administrators understand why a student may be at risk and what kind of support may be needed. This makes the system more useful in real-world academic environments, where decision-makers require both accuracy and interpretability. The project therefore contributes not only a predictive model, but also a decision-support framework that can assist in improving student retention and academic success.

Key achievements:

- Developed a machine learning-based dropout prediction model using relevant academic and socio-economic features
- Compared multiple algorithms to identify the most effective approach for the problem
- Selected the best model using F1-score to ensure balanced performance across precision and recall
- Handled class imbalance using SMOTE to improve minority class learning

- Built an interpretable system that provides risk levels, reasons, and suggestions for intervention

Overall, the proposed system shows that machine learning can play an important role in early dropout detection and student support. By enabling institutions to identify at-risk students before they disengage completely, the system can help improve retention rates, optimize counseling efforts, and support more informed academic decision-making. While the current implementation is effective, it also lays the foundation for future enhancements such as real-time deployment, integration with attendance and behavioral data, and testing on larger institutional datasets.

VI. FUTURE WORK

Although the proposed dropout risk detection system demonstrates strong predictive performance, there are several directions in which the work can be extended to improve accuracy, scalability, interpretability, and real-world applicability.

- Using advanced models like XGBoost or deep learning
  Future versions can explore stronger models such as XGBoost, LightGBM, CatBoost, and neural networks to capture more complex patterns and improve accuracy.
- Adding behavioral and attendance features
  The current system mainly uses academic and socio-economic data. Including attendance, assignment submission, and participation data may improve early dropout detection.
- Integration with real-time databases
  The system can be connected to student information systems and LMS platforms for automatic updates and continuous monitoring.
- Development of web/mobile applications
  A simple web or mobile interface can make the system easier to use for students, faculty, and administrators, with dashboards and alerts.
- Multi-institution dataset testing
  Testing the model on data from multiple institutions would help check whether it generalizes well across different student populations.
- Explainability and fairness analysis
  Techniques like SHAP or LIME can be added to explain predictions, while fairness checks can help reduce bias across demographic groups.
- Early intervention workflow support
  The system can also suggest actions such as counseling, mentoring, or fee support once a student is flagged as high risk.

Overall, future work should focus on making the system more accurate, explainable, and useful in real educational settings.

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